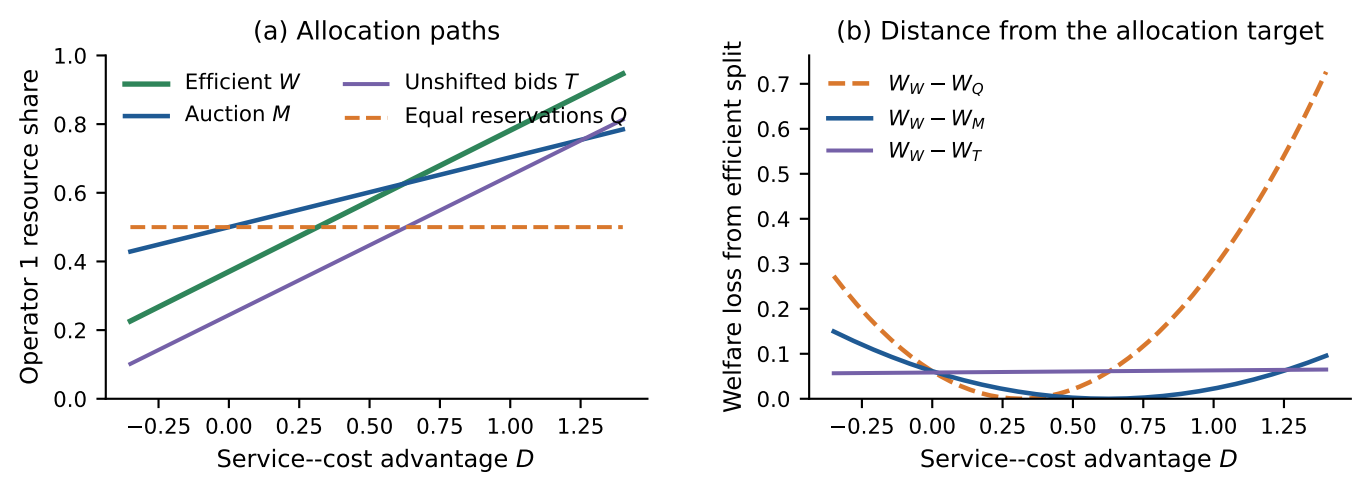
(a) Allocation paths
Operator 1 resource share
Service--cost advantage D
Efficient W
Auction M
Unshifted bids T
Equal reservations Q
1.0
0.8
0.6
0.4
0.2
0.0
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
(b) Distance from the allocation target
Welfare loss from efficient split
Service--cost advantage D
$W_W - W_Q$
$W_W - W_M$
$W_W - W_T$
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0